

On the probability that a binomial variable is at most its expectation

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ARTICLE INFO

Article history:

Received 16 October 2020

Received in revised form 4 December 2020

Accepted 6 December 2020

Available online 18 December 2020

MSC:

60C05

60E15

60F05

Keywords:

Binomial probabilities

Edgeworth expansions

Median

ABSTRACT

Consider the probability that a binomial random variable $\text{Bi}(n, m/n)$ with integer expectation m is at most its expectation. Chvátal conjectured that for any given n , this probability is smallest when m is the integer closest to $2n/3$. We show that this holds when n is large.

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1. Introduction

Consider the probability $\mathbb{P}(\text{Bi}(n, p) \leq np)$ that a binomial random variable $\text{Bi}(n, p)$ is less than or equal to its mean. (We slightly abuse notation, and let $\text{Bi}(n, p)$ denote both the binomial distribution and a binomial random variable.) By the central limit theorem, unless n or $p(1-p)$ is small, this probability is close to $\frac{1}{2}$; in fact, the Berry–Esseen theorem (Berry, 1941; Esseen, 1942) (see also e.g. Gut, 2013, Theorem 7.6.1) shows that $\mathbb{P}(\text{Bi}(n, p) \leq np) = \frac{1}{2} + O((np(1-p))^{-1/2})$. (See also the explicit bounds in Doerr, 2018; Greenberg and Mohri, 2014; Pelekis and Ramon, 2016; Rigollet and Tong, 2011; Slud, 1977.)

In the case when $np = m$ is an integer, Neumann (1966) showed that the mean np is also a (strong) median, i.e.,

$$\mathbb{P}(\text{Bi}(n, p) < np) < \frac{1}{2} < \mathbb{P}(\text{Bi}(n, p) \leq np), \quad np = m \in \{0, \dots, n\}. \quad (1.1)$$

(See Jogdeo and Samuels, 1968; Kaas and Buhrman, 1980, and Knuth, 2020, Exercise MPR-24 for other proofs.) It follows that for any fixed $n \geq 1$, the probability $\mathbb{P}(\text{Bi}(n, p) \leq np)$ regarded as a function of $p \in [0, 1]$, oscillates around $\frac{1}{2}$, with upward jumps at each m/n and monotone decrease between the jumps. See Fig. 1 for an example.

Consider again the case when $np = m$ is an integer, illustrated by the local maxima in Fig. 1. Vašek Chvátal (personal communication) made the following conjecture, based on numerical experiments.

Conjecture 1.1 (Chvátal). For any fixed $n \geq 2$, as m ranges over $\{0, \dots, n\}$, the probability $q_m := \mathbb{P}(\text{Bi}(n, m/n) \leq m)$ is smallest when m is the integer closest to $2n/3$.

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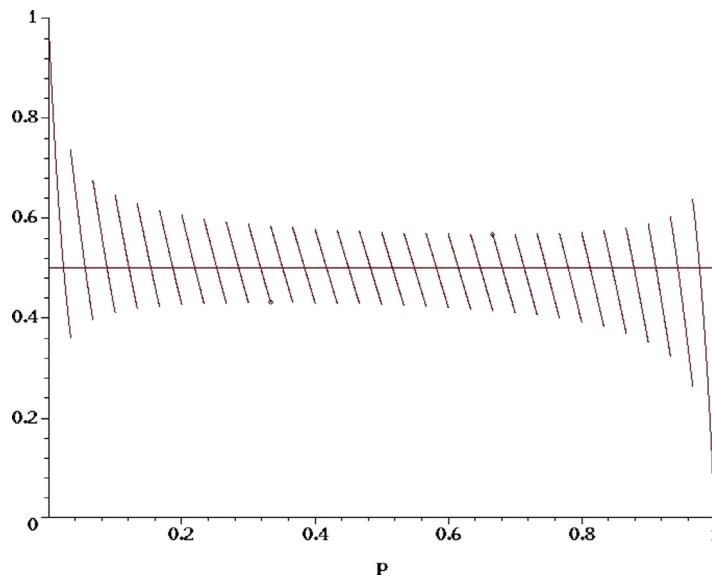


Fig. 1. $\mathbb{P}(\text{Bi}(n, p) \leq np)$ for $n = 30$. For integer values of $m = np$, the minimum of $\mathbb{P}(\text{Bi}(n, p) \leq np)$ (at $m = 20$) and the maximum of $\mathbb{P}(\text{Bi}(n, p) < np)$ (at $m = 10$) are marked with dots.

The purpose of the present paper is to show that this conjecture holds for large n . Moreover, at least for large n , the probabilities q_m are inverse unimodal, i.e., have no other local minimum. (The latter property was partly proved by [Rigollet and Tong, 2011](#), (29), who proved, for any n , that q_m decreases for $m \leq n/2$. We conjecture that also the inverse unimodality holds for all n .)

Theorem 1.2. *There exists n_0 such that [Conjecture 1.1](#) is true for every $n \geq n_0$. Moreover, still for $n \geq n_0$, the difference $q_{m+1} - q_m$ is negative when $m + \frac{1}{2} < 2n/3$ and positive when $m + \frac{1}{2} > 2n/3$.*

Remark 1.3. By symmetry, i.e., considering $n - \text{Bi}(n, m/n)$, it follows that for large n at least, the probability $\mathbb{P}(\text{Bi}(n, m/n) < m)$ is largest for the integer m closest to $n/3$. $\square$

Remark 1.4. For general p , the value of $\mathbb{P}(\text{Bi}(n, p) \leq np)$ is asymptotically given by

$$\mathbb{P}(\text{Bi}(n, p) \leq np) = \frac{1}{2} + \frac{4 - 2p - 6\{np\}}{6\sqrt{2\pi np(1-p)}} + O\left(\frac{1}{np(1-p)}\right), \quad (1.2)$$

at least provided $np(1-p) \geq \log^2 n$; this is a consequence of [Theorem 3.2](#) and (5.1) (with $k = 1$). Cf. [Fig. 1](#). See also the explicit related bound by [Doerr \(2018, Lemma 8\)](#). $\square$

Remark 1.5. In principle, it should be possible to calculate all constants in our proof explicitly, and thus find an explicit value for n_0 ; the conjecture then could be verified completely (assuming that it holds) by checking all smaller n numerically. However, we do not believe that this is practical. Presumably, other methods, completely different from ours, are needed to show Chvátal's conjecture in general. (We have verified the conjecture numerically for $n \leq 1000$.) $\square$

Our proof of [Theorem 1.2](#) is based on the version for integer-valued random variables found by [Esseen \(1945\)](#) of the asymptotic Edgeworth expansion for probabilities in the central limit theorem. This is usually stated for a single probability distribution, but we need to check that the estimates hold uniformly for $\text{Bi}(n, p)$ with p in some range; hence we discuss this expansion in some detail in [Section 3](#). In particular, we state in [Theorem 3.2](#) the result that we need in a general form, and prove it in [Section 4](#). We return to the binomial probabilities in [Section 5](#), and prove [Theorem 1.2](#) in [Section 6](#).

Remark 1.6. See [Brown et al. \(2002\)](#) for another aspect, with statistical implications, of the oscillations of binomial probabilities; they too use Edgeworth expansions with more than one term. $\square$

2. Preliminaries

2.1. General notation

We let $\{x\} := x - \lfloor x \rfloor$ denote the fractional part of a real number x ; thus $\{x\} \in [0, 1)$.

$f^{(m)}$ denotes the m th derivative of a function f , with $f^{(0)} := f$.

A random variable X has *span* d if it is concentrated on a set $x_0 + d\mathbb{Z}$ for some real x_0 , and d is maximal with this property.

C and c denote unimportant constants, in general different at different occurrences. The “constants” may depend on some given parameters, given by the context; we sometimes write e.g. C_k to emphasize that C depends on a parameter k , but this is omitted when not necessary.

2.2. Special notation

We introduce here the notation needed for the expansions in the following sections. See further [Esseen \(1945, Chapters III–IV\)](#) and [Petrov \(1975, §VI.1\)](#).

Let X be a random variable. (X will be regarded as given. Most quantities defined below depend on the distribution of X , although we for simplicity do not show this in the notation.) Denote its mean by $\mu := \mathbb{E}X$, its central absolute moments by $\beta_j := \mathbb{E}|X - \mu|^j$, and its cumulants by γ_j (when they exist, i.e., when $\mathbb{E}|X|^j < \infty$). Also, let $\sigma^2 = \beta_2 = \gamma_2 = \text{Var } X$ be the variance of X , and define the scale-invariant

$$\Lambda_j := \frac{\beta_j}{\sigma^j}, \quad (2.1)$$

$$\lambda_j := \frac{\gamma_j}{\sigma^j}. \quad (2.2)$$

Each cumulant $\gamma_j, j \geq 2$, can be expressed as a polynomial in central moments of orders $2, \dots, j$, and it follows, using also Hölder's inequality, that

$$|\gamma_j| \leq C_j \beta_j \quad (2.3)$$

and thus

$$|\lambda_j| \leq C_j \Lambda_j. \quad (2.4)$$

Furthermore, Hölder's inequality also easily yields, for $2 \leq j \leq k$,

$$1 = \Lambda_2 \leq \Lambda_j^{1/(j-2)} \leq \Lambda_k^{1/(k-2)}. \quad (2.5)$$

Define polynomials $P_j(u)$, $j \geq 1$, by expanding the formal power series

$$\exp\left(\sum_{i=1}^{\infty} \frac{\lambda_{i+2}}{(i+2)!} u^{i+2} z^i\right) = 1 + \sum_{j=1}^{\infty} P_j(u) z^j. \quad (2.6)$$

Note that $P_j(u)$ is a polynomial of degree $3j$; moreover,

$$P_j(u) = \sum_{r=j+2}^{3j} \varpi_{jr} u^r, \quad (2.7)$$

where each coefficient ϖ_{jr} is a polynomial in $\lambda_k, k = 3, \dots, j+2$. In particular, $P_j(u)$ is well-defined provided $\mathbb{E}|X|^{j+2} < \infty$. Furthermore, $\varpi_{jr} = 0$ unless $r - j$ is even.

Let $\Phi(x)$ be the distribution function of the standard normal distribution, so that

$$\frac{d}{dx} \Phi(x) = \phi(x) := \frac{1}{\sqrt{2\pi}} e^{-x^2/2}. \quad (2.8)$$

Define $Q_j(x)$ as the function obtained from $P_j(u)$ by replacing each power u^r by $(-1)^r \Phi^{(r)}(x) = (-1)^r \frac{d^r}{dx^r} \Phi(x)$, i.e.,

$$Q_j(x) := \sum_{r=j+2}^{3j} (-1)^r \varpi_{jr} \Phi^{(r)}(x) = \sum_{r=j+2}^{3j} (-1)^r \varpi_{jr} \phi^{(r-1)}(x) \quad (2.9)$$

$$= -\phi(x) \sum_{r=j+2}^{3j} \varpi_{jr} H_{r-1}(x), \quad (2.10)$$

where $H_r(x)$ are the Hermite polynomials (in the normalization natural in probability theory, i.e., orthogonal w.r.t. the standard normal distribution); see e.g. [Olver et al. \(2010, \(18.5.9\)\)](#) (there denoted $He_r(x)$) or [Petrov \(1975, p. 127\)](#).

Define also periodic functions ψ_r , $r \geq 1$, by their Fourier series

$$\psi_r(x) = \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} \frac{e^{2\pi kxi}}{(2\pi ki)^r}. \quad (2.11)$$

Note that for $r \geq 2$, the series (2.11) converges absolutely and defines a continuous periodic function with period 1. However, for $r = 1$, the series is only conditionally convergent; in fact it is the Fourier series of $\frac{1}{2} - \{x\}$. (It follows from standard results that the series converges for every x , but we do not really need this.) Hence, $\psi_1(x)$ has a jump 1 at every integer. For later convenience, we redefine ψ_1 to be right-continuous; thus we define

$$\psi_1(x) := \frac{1}{2} - \{x\} = \frac{1}{2} - x + \lfloor x \rfloor, \quad x \in \mathbb{R}, \quad (2.12)$$

noting that for $r = 1$, (2.11) holds only for non-integer x . Note also that, for any $r \geq 1$ and $x \in \mathbb{R}$,

$$\psi_r(x) = -\frac{1}{r!} B_r(\{x\}), \quad (2.13)$$

where $B_r(x)$ denotes the Bernoulli polynomials, see e.g. Olver et al. (2010, (24.8.3)). In particular, Olver et al. (2010, (24.2.4)),

$$\psi_r(0) = -\frac{1}{r!} B_r(0) = -\frac{1}{r!} B_r, \quad (2.14)$$

where B_r denotes the Bernoulli numbers. Recall that $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$ and $B_{2j+1} = 0$ for $j \geq 1$.

Remark 2.1. Note that $\psi_r(x) = (-1)^{r-1} h_r Q_r(x)$ (where $h_r = \pm 1$) in the notation of Esseen (1945, p. 60–61). We prefer the choice of signs in our definition, but this is only a matter of taste. $\square$

3. The basic expansion theorem

Esseen (1945, Theorem IV.4) proved the following. More precisely, Esseen's result is (3.4); the version (3.5) follows immediately by applying (3.4) with $k + 1$ instead of k . (We prefer this, weaker, version for our generalization in Theorem 3.2.)

Theorem 3.1 (Esseen, 1945). *Let $X, X_1, X_2, \dots$ be i.i.d. integer valued random variables with span 1 and let $S_n := \sum_{i=1}^n X_i$. Let $k \geq 1$ be an integer and suppose that $\mathbb{E}|X|^{k+2} < \infty$. With notations as above, define*

$$\Psi_{n,k}(x) := \Phi(x) + \sum_{j=1}^k n^{-j/2} Q_j(x) \quad (3.1)$$

and

$$\Psi_{n,k}^*(x) := \Psi_{n,k}(x) + \sum_{\ell=1}^k (-1)^{\ell-1} n^{-\ell/2} \sigma^{-\ell} \psi_\ell(n\mu + x\sigma\sqrt{n}) \Psi_{n,k}^{(\ell)}(x). \quad (3.2)$$

Then

$$\mathbb{P}(S_n \leq n\mu + x\sigma\sqrt{n}) = \Psi_{n,k}^*(x) + R_{n,k}(x), \quad (3.3)$$

where, as $n \rightarrow \infty$,

$$\sup_x |R_{n,k}(x)| = o(n^{-k/2}). \quad (3.4)$$

If further $\mathbb{E}|X|^{k+3} < \infty$, then

$$\sup_x |R_{n,k}(x)| = O(n^{-\frac{k+1}{2}}). \quad (3.5)$$

Theorem 3.1 is stated for a single distribution. We want to apply it to $X \sim \text{Be}(p)$, but then need some uniform estimates for all p , or at least for a large range. It is no surprise that the proof of Theorem 3.1 yields such uniformity under suitable conditions, including some uniform moment estimates. For $\text{Be}(p)$, the case $p \in [p_1, p_2]$ for a compact interval $[p_1, p_2] \subset (0, 1)$ does not cause any difficulties, but we can go beyond that. We will show the following extension of Theorem 3.1 in Section 4.

Theorem 3.2. Let $k \geq 1$. For every $c > 0$, there exists a constant C (depending on k and c only) such that if X is an integer-valued random variable with $\mathbb{E}|X|^{k+3} < \infty$ and

$$\min\{\mathbb{P}(X=0), \mathbb{P}(X=1)\} \geq c\sigma^2, \quad (3.6)$$

and $n \geq 2$ is an integer with

$$\sigma\sqrt{n} \geq \log n, \quad (3.7)$$

then (3.3) holds with

$$\sup_x |R_{n,k}(x)| \leq C\Lambda_{k+3}n^{-\frac{k+1}{2}} \leq C\frac{\mathbb{E}|X|^{k+3}}{\sigma^{k+3}}n^{-\frac{k+1}{2}}. \quad (3.8)$$

Remark 3.3. We defined $\psi_1(x)$ to be right-continuous so that $\Psi_{n,k}^*(x)$ and $R_{n,k}(x)$ are right-continuous, which enables (3.4), (3.5) and (3.8) to hold for all x , including integers. (All functions in (3.1)–(3.2) except ψ_1 are continuous.) $\square$

Remark 3.4. As remarked by Esseen (1945, p. 61), (3.2) contains redundant terms. The form (3.2) is sometimes convenient (for example in the proof), but it is often more convenient to modify (3.2) by dropping the redundant terms; we thus define also

$$\tilde{\Psi}_{n,k}^*(x) := \Psi_{n,k}(x) + \sum_{\ell=1}^k (-1)^{\ell-1} n^{-\ell/2} \sigma^{-\ell} \psi_\ell(n\mu + x\sigma\sqrt{n}) \Psi_{n,k-\ell}^{(\ell)}(x) \quad (3.9)$$

and define a modified remainder term $\tilde{R}_{n,k}(x)$ by

$$\mathbb{P}(S_n \leq n\mu + x\sigma\sqrt{n}) = \tilde{\Psi}_{n,k}^*(x) + \tilde{R}_{n,k}(x). \quad (3.10)$$

It follows from (3.1) that in Theorem 3.1, this changes $\Psi_{n,k}^*(x)$ and thus $R_{n,k}(x)$ by some terms which are $O(n^{-m/2})$ for some $m \geq k+1$ so (3.4) and (3.5) still hold for $\tilde{R}_{n,k}(x)$. With only a little more effort, it can be verified that the same holds for Theorem 3.2; it follows from Lemma 4.1 that the removed terms are all dominated by $\Lambda_{k+3}n^{-\frac{k+1}{2}}$, so (3.8) still holds. (This uses also that $\sigma\sqrt{n}$ is bounded below by the assumptions, and the fact that $\beta_i \leq C_{i,j}\beta_j$ when $2 \leq i \leq j$ and X is integer valued.) $\square$

Remark 3.5. Note that Theorem 1.2 is only for random variables with span 1, and that a uniform version for a family of random variables therefore requires some uniform condition preventing the variables from being too close to variables with larger span. We use condition (3.6) which is convenient and turns out to be sufficient; it can obviously be replaced by more general conditions. $\square$

Remark 3.6. The assumption (3.7) in Theorem 3.2 is annoying but not a very serious restriction. Note that the right-hand side of (3.8) is, since X is integer-valued, at least $C(\sigma\sqrt{n})^{-k-1}$. Hence, if (3.7) is violated, (3.8) would, even if true, only give a weak bound. We do not know whether (3.7) really is needed. It is possible that Theorem 3.2 could be proved without this assumption, using the alternative method of proof in Esseen (1945, Section IV.4), but we have not pursued this. $\square$

4. Proof of Theorem 3.2

Lemma 4.1. Suppose that $\ell \geq 1$ and $\mathbb{E}|X|^{\ell+2} < \infty$. Then, for every $m \geq 0$,

$$|\varpi_{\ell,r}| \leq C_\ell \Lambda_{\ell+2}, \quad (4.1)$$

$$|P_\ell(u)| \leq C_\ell \Lambda_{\ell+2}(|u|^{\ell+2} + |u|^{3\ell}), \quad (4.2)$$

$$|Q_\ell^{(m)}(x)| \leq C_{\ell,m} \Lambda_{\ell+2}, \quad (4.3)$$

$$|\Psi_{n,\ell}^{(m)}(x)| \leq C_{\ell,m}(1 + \Lambda_{\ell+2}n^{-\ell/2}). \quad (4.4)$$

Proof. It follows from (2.6) that $\varpi_{\ell,r}$ is a linear combination of products $\prod_{k=1}^m \lambda_{i_k+2}$ with $\sum_k(i_k+2) = r$ and $\sum_k i_k = \ell$. By (2.4) and (2.5), each such product is bounded by

$$C_\ell \prod_{k=1}^m \Lambda_{i_k+2} \leq C_\ell \prod_{k=1}^m \Lambda_{\ell+2}^{i_k/\ell} = C_\ell \Lambda_{\ell+2}, \quad (4.5)$$

which yields (4.1). This implies (4.2) and (4.3) by (2.7) and (2.9), noting that $\Phi(x)$ and all its derivatives are bounded on $\mathbb{R}$. Finally, (3.1) and (4.3) together with (2.5) yield

$$\begin{aligned} |\Psi_{n,\ell}^{(m)}(x)| &\leq C_m + C_{\ell,m} \sum_{j=1}^{\ell} n^{-j/2} \Lambda_{j+2} \leq C_{\ell,m} \sum_{j=0}^{\ell} n^{-j/2} \Lambda_{\ell+2}^{j/\ell} \\ &\leq C_{\ell,m} (1 + n^{-\ell/2} \Lambda_{\ell+2}). \quad \square \end{aligned} \quad (4.6)$$

Lemma 4.2. If X is a random variable with $\mathbb{P}(X = 0) \geq a$ and $\mathbb{P}(X = 1) \geq a$ for some $a \geq 0$, then

$$|\mathbb{E}e^{itX}| \leq 1 - \frac{a}{\pi^2} t^2 \leq e^{-a\pi^{-2}t^2}, \quad |t| \leq \pi. \quad (4.7)$$

Proof. The assumption implies that $\mathbb{E}e^{itX} - a - ae^{it}$ is the Fourier transform of a positive measure with mass $1 - 2a$. Hence, if $|t| \leq \pi$,

$$\begin{aligned} |\mathbb{E}e^{itX}| &\leq |a + ae^{it}| + |\mathbb{E}e^{itX} - a - ae^{it}| \leq a|1 + e^{it}| + 1 - 2a \\ &= 2a \left| \cos \frac{t}{2} \right| + 1 - 2a = 1 - 2a(1 - \cos \frac{t}{2}) = 1 - 4a \sin^2 \frac{t}{4}, \end{aligned} \quad (4.8)$$

and (4.7) follows. $\square$

Proof of Theorem 3.2. We follow the proofs of Theorems 3 and 4 in Esseen (1945) (with $d = 1$ and thus $t_0 = 2\pi$) and mention only the main differences. Note that Esseen (1945) considers centred variables, so X_i there is our $X_i - \mu$. Let

$$f(t) := \mathbb{E}e^{it(X-\mu)}, \quad (4.9)$$

$$f_n(t) := \mathbb{E}e^{it(S_n - n\mu)/(\sigma\sqrt{n})} = f^n(t/(\sigma\sqrt{n})), \quad (4.10)$$

$$g_0(t) := e^{-t^2/2} + \sum_{j=1}^k n^{-j/2} P_j(it) e^{-t^2/2}. \quad (4.11)$$

Also, let $T := n^{\frac{k+1}{2}}$ and replace T_{3n} in Esseen (1945) by

$$T_1 := \sqrt{n}/\Lambda_{k+3}^{1/(k+1)}. \quad (4.12)$$

The second inequality in (3.8) is trivial, by the definition (2.1). We may also assume that $\Lambda_{k+3} n^{-\frac{k+1}{2}} \leq 1$, and thus $T_1 \geq 1$, since otherwise it follows from Lemma 4.1 and (2.5) that each term in (3.2) is bounded by $C\Lambda_{k+3} n^{-\frac{k+1}{2}}$, and thus (3.8) holds trivially.

In the range $|t| \leq T_1$, we have

$$|f_n(t) - g_0(t)| \leq C\Lambda_{k+3}|t|^{k+3}e^{-t^2/13}n^{-\frac{k+1}{2}} \quad (4.13)$$

by Petrov (1975, Lemma VI.4) with $s = k+3$ and $m = 0$ (which essentially is Esseen, 1945, Lemma III.2a with T_{3n} improved to our $T_1/4$, which can be proved in the same way). Hence, for the “main term” in the estimate

$$\int_{-T_1}^{T_1} \frac{|f_n(t) - g_0(t)|}{|t|} dt \leq C\Lambda_{k+3}n^{-\frac{k+1}{2}}. \quad (4.14)$$

Furthermore, if $|t| \leq \pi\sigma\sqrt{n}$, then the assumption (3.6) and Lemma 4.2 yield

$$|f_n(t)| = |f(t/(\sigma\sqrt{n}))|^n \leq e^{-nc\sigma^2(t/\sigma\sqrt{n})^2} = e^{-ct^2}. \quad (4.15)$$

Hence,

$$\int_{T_1}^{\pi\sigma\sqrt{n}} \frac{|f_n(t)|}{|t|} dt \leq \int_{T_1}^{\pi\sigma\sqrt{n}} |f_n(t)| dt \leq Ce^{-cT_1^2} \leq CT_1^{-(k+1)} = C\Lambda_{k+3}n^{-\frac{k+1}{2}}. \quad (4.16)$$

The integral $\int_{T_1}^{\pi\sigma\sqrt{n}} |g_0(t)|/|t| dt$ has the same estimate. Consequently, by (4.14) and (4.16),

$$\int_{-\pi\sigma\sqrt{n}}^{\pi\sigma\sqrt{n}} \frac{|f_n(t) - g_0(t)|}{|t|} dt \leq C\Lambda_{k+3}n^{-\frac{k+1}{2}}. \quad (4.17)$$

The same arguments yield also

$$\int_{-\pi\sigma\sqrt{n}}^{\pi\sigma\sqrt{n}} |f_n(t) - g_0(t)| dt \leq C\Lambda_{k+3}n^{-\frac{k+1}{2}}. \quad (4.18)$$

Using the estimates (4.17) and (4.18), the rest of the proof is essentially the same as in Esseen (1945). One of the terms, generalizing I'_k on Esseen (1945, p. 58–59), is

$$\int_{-\pi\sigma\sqrt{n}}^{\pi\sigma\sqrt{n}} \frac{|f_n(t) - g_0(t)|}{|t + 2\pi j\sigma\sqrt{n}|} dt \leq \frac{C}{|j|\sigma\sqrt{n}} \Lambda_{k+3} n^{-\frac{k+1}{2}}, \quad (4.19)$$

where we use (4.18). This term exists for all integers $j \neq 0$ with $(2|j| - 1)\pi\sigma\sqrt{n} < T = n^{\frac{k+1}{2}}$, and thus the sum of them is bounded by

$$2 \sum_{j=1}^T \frac{C}{|j|\sigma\sqrt{n}} \Lambda_{k+3} n^{-\frac{k+1}{2}} \leq C \frac{\log n}{\sigma\sqrt{n}} \Lambda_{k+3} n^{-\frac{k+1}{2}}; \quad (4.20)$$

this is the reason for our assumption $\log n \leq \sigma\sqrt{n}$, which leads to an estimate $C \Lambda_{k+3} n^{-\frac{k+1}{2}}$ for (4.19) too. The remaining terms give no problems. $\square$

5. Expansions for the binomial probabilities

Taking $x = 0$ in (3.9)–(3.10), we obtain

$$\mathbb{P}(S_n \leq n\mu) = \psi_{n,k}(0) + \sum_{\ell=1}^k (-1)^{\ell-1} n^{-\ell/2} \sigma^{-\ell} \psi_{\ell}(n\mu) \psi_{n,k-\ell}^{(\ell)}(0) + \tilde{R}_{n,k}(0). \quad (5.1)$$

If furthermore $n\mu$ is an integer, this yields, using (2.14) and $B_0 := 1$, and for convenience defining $Q_0(x) := \Phi(x)$,

$$\begin{aligned} \mathbb{P}(S_n \leq n\mu) &= \psi_{n,k}(0) + \sum_{\ell=1}^k (-1)^{\ell-1} n^{-\ell/2} \sigma^{-\ell} \psi_{\ell}(0) \psi_{n,k-\ell}^{(\ell)}(0) + \tilde{R}_{n,k}(0) \\ &= \sum_{\ell=0}^k \sum_{j=0}^{k-\ell} n^{-(j+\ell)/2} \sigma^{-\ell} \frac{(-1)^{\ell} B_{\ell}}{\ell!} Q_j^{(\ell)}(0) + \tilde{R}_{n,k}(0) \\ &= \frac{1}{2} + \sum_{m=1}^k n^{-m/2} \sum_{\ell=0}^m \sigma^{-\ell} \frac{(-1)^{\ell} B_{\ell}}{\ell!} Q_{m-\ell}^{(\ell)}(0) + \tilde{R}_{n,k}(0). \end{aligned} \quad (5.2)$$

Since $\phi(x)$ is an even function, all its odd derivatives vanish at 0, and since $\varpi_{jr} = 0$ unless $j \equiv r \pmod{2}$, it follows from (2.9) that $Q_j^{(\ell)}(0) = 0$ when $j \equiv \ell \pmod{2}$ (except when $j = \ell = 0$). Hence, all terms in (5.2) with m even vanish.

Remark 5.1. For $\mathbb{P}(S_n < n\mu)$, we have the same formulas with $\psi_1(x)$ replaced by its left-continuous version $\psi_1(x-)$. (All other appearing functions are continuous.) In (5.2), this means that the sign is changed for the terms with $\ell = 1$; all other terms remain the same. $\square$

We specialize to $X \sim \text{Be}(p)$, with $0 < p < 1$, so $S_n \sim \text{Bi}(n, p)$. All moments exist, and we have $\sigma^2 := p(1-p)$. Furthermore, $\beta_j := \mathbb{E}|X - p|^j \leq \beta_2 = \sigma^2$; hence, recalling (2.1) and (2.4),

$$|\lambda_j| \leq C_j \Lambda_j \leq C_j \sigma^{2-j}. \quad (5.3)$$

The condition (3.6) holds with $c = 1$ for all p . Hence Theorem 3.2 applies provided

$$\sqrt{np(1-p)} \geq \log n, \quad (5.4)$$

and then yields, together with Remark 3.4 and using (5.3),

$$|R_{n,k}(x)|, |\tilde{R}_{n,k}(x)| \leq C_k \Lambda_{k+3} n^{-\frac{k+1}{2}} \leq \frac{C_k}{(n\sigma^2)^{(k+1)/2}} = \frac{C_k}{(np(1-p))^{(k+1)/2}} \quad (5.5)$$

We use (5.2) with $k = 4$ and obtain, when np is an integer,

$$\mathbb{P}(S_n \leq np) = \frac{1}{2} + h_1(p)n^{-1/2} + h_3(p)n^{-3/2} + O((np(1-p))^{-5/2}), \quad (5.6)$$

where, after some calculations using (2.10) and (2.6) after finding the cumulants $\gamma_3, \gamma_4, \gamma_5$,

$$h_1(p) := Q_1(0) + \frac{1}{2\sigma} Q'_0(0) = \frac{\gamma_3}{6\sqrt{2\pi}\sigma^3} + \frac{1}{2\sqrt{2\pi}\sigma} = \frac{2-p}{3\sqrt{2\pi}\sqrt{p(1-p)}}, \quad (5.7)$$

$$h_3(p) := Q_3(0) + \frac{1}{2\sigma} Q'_2(0) + \frac{1}{12\sigma^2} Q''_1(0) = \frac{(2-p)(p^2 + 23p - 23)}{540\sqrt{2\pi}(p(1-p))^{3/2}}. \quad (5.8)$$

Recall that there is no $h_2(p)$ or $h_4(p)$; the corresponding terms vanish as said above.

6. Proof of Theorem 1.2

We now prove our main result.

Proof of Theorem 1.2. The main idea of the proof is to estimate $q_{m+1} - q_m$ using the estimates above, in particular (5.6), but the details will differ for different ranges of m . We write $p_m := m/n$. We sometimes tacitly assume that n is large enough. C_1, C_2, C_3 will denote some large constants.

Recall $h_1(p)$ and $h_3(p)$ given in (5.7)–(5.8). A simple differentiation yields

$$h'_1(p) = \frac{3p - 2}{6\sqrt{2\pi}(p(1-p))^{3/2}}. \quad (6.1)$$

Note that $h'_1(p) = 0$ for $p = 2/3$, with $h'(p) < 0$ for $0 < p < 2/3$ and $h'(p) > 0$ for $2/3 < p < 1$; this is the fundamental reason for the behaviour shown in Theorem 1.2, although we also have to treat error terms.

There is no need to calculate $h'_3(p)$ exactly; it suffices to note from (5.8) that

$$h'_3(p) = O((p(1-p))^{-5/2}). \quad (6.2)$$

We treat several cases separately.

Case 1: $2 \log^2 n \leq m \leq n - 2 \log^2 n$. Both p_m and p_{m+1} satisfy (5.4); hence Theorem 3.2 applies and yields (5.6) for both. By subtraction and the mean value theorem, we obtain, for some $p'_m, p''_m \in [p_m, p_{m+1}]$, recalling that $p_{m+1} - p_m = n^{-1}$ and using (6.1)–(6.2),

$$q_{m+1} - q_m = h'_1(p'_m)n^{-3/2} + h'_3(p''_m)n^{-5/2} + O((np_m(1-p_m))^{-5/2}) \quad (6.3)$$

$$= \frac{3p_m - 2 + O(1/n)}{6\sqrt{2\pi}}(np'_m(1-p'_m))^{-3/2} + O((np_m(1-p_m))^{-5/2}). \quad (6.4)$$

Case 1a: $2 \log^2 n \leq m \leq n/2$. In this subcase, (6.4) yields

$$q_{m+1} - q_m \leq -c(np_m)^{-3/2} + O((np_m)^{-5/2}) < 0, \quad (6.5)$$

provided n and thus $np_m = m$ is large enough.

Case 1b: $n/2 \leq m \leq 2n/3 - C_1$. In this subcase, (6.4) similarly yields

$$q_{m+1} - q_m \leq -\frac{3C_1}{6\sqrt{2\pi}}cn^{-5/2} + O(n^{-5/2}) < 0, \quad (6.6)$$

provided C_1 is chosen large enough.

Case 1c: $2n/3 + C_2 \leq m \leq n - 2 \log^2 n$. Similar arguments as in the two preceding subcases yield $q_{m+1} - q_m > 0$.

Case 1d: $2n/3 - C_1 \leq m \leq 2n/3 + C_2$. This is the most delicate case, since $p_m - 2/3 = O(1/n)$ and the three terms in (6.3) all are of the same order. We thus expand one step further and use Theorem 3.2 with $k = 6$; this yields, again using (5.2),

$$q_m = \frac{1}{2} + h_1(p_m)n^{-1/2} + h_3(p_m)n^{-3/2} + h_5(p_m)n^{-5/2} + O(n^{-7/2}), \quad (6.7)$$

where we note that $h_5(p)$ is a differentiable function of p . (It can easily be calculated, but we do not need this.) Taylor expansions yield, recalling $h'_1(\frac{2}{3}) = 0$,

$$\begin{aligned} q_m &= \frac{1}{2} + h_1\left(\frac{2}{3}\right)n^{-1/2} + \frac{1}{2}h''_1\left(\frac{2}{3}\right)\left(p_m - \frac{2}{3}\right)^2n^{-1/2} + h_3\left(\frac{2}{3}\right)n^{-3/2} \\ &\quad + h'_3\left(\frac{2}{3}\right)\left(p_m - \frac{2}{3}\right)n^{-3/2} + h_5\left(\frac{2}{3}\right)n^{-5/2} + O(n^{-7/2}) \\ &= G(n) + H\left(m - 2n/3\right)n^{-5/2} + O(n^{-7/2}), \end{aligned} \quad (6.8)$$

where we define

$$G(n) := \frac{1}{2} + h_1\left(\frac{2}{3}\right)n^{-1/2} + h_3\left(\frac{2}{3}\right)n^{-3/2} + h_5\left(\frac{2}{3}\right)n^{-5/2}, \quad (6.9)$$

$$H(x) := \frac{1}{2}h''_1\left(\frac{2}{3}\right)x^2 + h'_3\left(\frac{2}{3}\right)x. \quad (6.10)$$

The formula (6.8) holds for $m + 1$ too, and thus

$$\begin{aligned} q_{m+1} - q_m &= (H(m + 1 - 2n/3) - H(m - 2n/3))n^{-5/2} + O(n^{-7/2}) \\ &= \left(h''_1\left(\frac{2}{3}\right)\left(m - \frac{2n}{3} + \frac{1}{2}\right) + h'_3\left(\frac{2}{3}\right)\right)n^{-5/2} + O(n^{-7/2}) \end{aligned} \quad (6.11)$$

A calculation yields

$$h_1''(2/3) = \frac{27}{8\sqrt{\pi}}, \quad (6.12)$$

$$h_3'(2/3) = \frac{9}{40\sqrt{\pi}} = \frac{1}{15}h_1''(2/3). \quad (6.13)$$

Since the ratio $1/15 < 1/6$, it follows from (6.11) that if $m \leq (2n-2)/3$, then $q_{m+1} - q_m < 0$, and if $m \geq (2n-1)/3$, then $q_{m+1} - q_m > 0$, for large n .

Case 2: $m < 2 \log^2 n$. As said in the introduction, Rigollet and Tong (2011, (29)) showed that (for every n) $q_{m-1} \geq q_m$, and their proof actually gives $q_{m-1} > q_m$, for $m \leq n/2$. Alternatively (for large n), we can argue using Poisson approximation as in the next case; we omit the details.

Case 3: $m > n - 2 \log^2 n$. Define $q'_m := \mathbb{P}(\text{Bi}(n, m/n) < m)$. By symmetry, $1 - q_m = q'_{n-m}$; hence the claim $q_m < q_{m+1}$ is equivalent to $q'_{m-1} < q'_m$ for $m < 2 \log^2 n$.

We use Poisson approximation of the binomial distribution. It is well-known, see e.g. Barbour et al. (1992, Theorem 2.M) that the total variation distance between $\text{Bi}(n, p)$ and $\text{Po}(np)$ is less than p , and thus, in particular,

$$|q'_m - \mathbb{P}(\text{Po}(m) < m)| \leq d_{\text{TV}}(\text{Bi}(n, p_m), \text{Po}(np_m)) < p_m. \quad (6.14)$$

We estimate $\mathbb{P}(\text{Po}(m) < m)$ by Theorem 3.2 (or Theorem 3.1) applied to $X \sim \text{Po}(1)$. This yields, using (5.2) and Remark 5.1,

$$\mathbb{P}(\text{Po}(m) < m) = \frac{1}{2} - \frac{1}{3\sqrt{2\pi}}m^{-1/2} - \frac{23}{270\sqrt{2\pi}}m^{-3/2} + O(m^{-5/2}). \quad (6.15)$$

Combining (6.14) and (6.15) we find, for $m \geq 2$,

$$q'_m - q'_{m-1} = \frac{1}{6\sqrt{2\pi}}m^{-3/2} + O(m^{-5/2}) + O(m/n). \quad (6.16)$$

This shows that $q'_m > q'_{m-1}$ for $C_3 < m < 2 \log^2 n$ and n large. (In fact, for $C_3 < m < cn^{2/5}$.)

In the remaining subcase $m \leq C_3$, $q'_{m-1} < q'_m$ follows from (6.14) and Lemma 6.1.

These cases cover all m , which completes the proof of Theorem 1.2. $\square$

The proof used the following lemma, of independent interest. It gives two Poisson versions of the inequality mentioned above for the binomial distribution shown by Rigollet and Tong (2011). We use their method of proof.

Lemma 6.1. For every integer $m \geq 0$,

$$\mathbb{P}(\text{Po}(m) < m) < \mathbb{P}(\text{Po}(m+1) < m+1) < \frac{1}{2}, \quad (6.17)$$

$$\mathbb{P}(\text{Po}(m) \leq m) > \mathbb{P}(\text{Po}(m+1) \leq m+1) > \frac{1}{2}. \quad (6.18)$$

Proof. We may assume $m \geq 1$. Consider a Poisson process with intensity 1 on $(0, \infty)$ and denote its points by $T_1 < T_2 < \dots$. The number of points in $[0, m]$ is $\text{Po}(m)$, and thus $\text{Po}(m) < m \iff T_m > m$. The distribution of T_m is $\Gamma(m)$, and thus

$$\mathbb{P}(\text{Po}(m) < m) = \mathbb{P}(T_m > m) = \int_m^\infty \frac{t^{m-1}}{(m-1)!} e^{-t} dt. \quad (6.19)$$

Hence,

$$\begin{aligned} & \mathbb{P}(\text{Po}(m) < m) - \mathbb{P}(\text{Po}(m+1) < m+1) \\ &= \int_m^\infty \frac{t^{m-1}}{(m-1)!} e^{-t} dt - \int_{m+1}^\infty \frac{t^m}{m!} e^{-t} dt \\ &= \int_m^\infty \left(\frac{t^{m-1}}{(m-1)!} e^{-t} - \frac{t^m}{m!} e^{-t} \right) dt + \int_m^{m+1} \frac{t^m}{m!} e^{-t} dt \\ &= \left[\frac{t^m}{m!} e^{-t} \right]_m^\infty + \int_m^{m+1} \frac{t^m}{m!} e^{-t} dt = \int_m^{m+1} \frac{t^m e^{-t} - m^m e^{-m}}{m!} dt < 0, \end{aligned} \quad (6.20)$$

since $t^m e^{-t}$ is decreasing for $t \geq m$. Similarly,

$$\begin{aligned} & \mathbb{P}(\text{Po}(m) \leq m) - \mathbb{P}(\text{Po}(m+1) \leq m+1) \\ &= \int_m^\infty \frac{t^m}{m!} e^{-t} dt - \int_{m+1}^\infty \frac{t^{m+1}}{(m+1)!} e^{-t} dt \end{aligned}$$

$$\begin{aligned}
&= \int_m^\infty \left(\frac{t^m}{m!} e^{-t} - \frac{t^{m+1}}{(m+1)!} e^{-t} \right) dt + \int_m^{m+1} \frac{t^{m+1}}{(m+1)!} e^{-t} dt \\
&= \int_m^{m+1} \frac{t^{m+1} e^{-t} - m^{m+1} e^{-m}}{(m+1)!} dt > 0,
\end{aligned} \tag{6.21}$$

since $t^{m+1}e^{-t}$ is increasing for $t \leq m+1$. $\square$

Acknowledgements

This work is dedicated to the memory of my teacher and colleague Carl-Gustav Esseen (1918–2001). I thank Vašek Chvátal for introducing me to this problem, and Xing Shi Cai and an anonymous referee for finding minor errors in a previous version.

Supported by the Knut and Alice Wallenberg Foundation, Sweden.

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